1 Data Generation

1.1 Pattern Matching

A *pattern* is a set.

1.1.1 Example

Let $a * b = [a_1, \cdots, a_n, b_1, \cdots, b_m]$, where $a = [a_1, \cdots, a_n]$ and $b = [b_1, \cdots, b_m]$.

$$
\begin{array}{lcl}
\mathfrak{E} & := & \mathfrak{T} \cup \{\mathfrak{e}_1 * \mathfrak{o} * \mathfrak{e}_2 \,|\, \mathfrak{o} \in \{[\text{“}+\text{”}], [\text{“}-\text{”}]\} \wedge \mathfrak{e}_1, \mathfrak{e}_2 \in \mathfrak{E}\} \\
\mathfrak{T} & := & \mathfrak{F} \cup \{\mathfrak{t}_1 * \mathfrak{o} * \mathfrak{t}_2 \,|\, \mathfrak{o} \in \{[\text{“}\times\text{”}], [\text{“}\div\text{”}]\} \wedge \mathfrak{t}_1, \mathfrak{t}_2 \in \mathfrak{T}\} \\
\mathfrak{F} & := & \mathfrak{N} \cup \{[\text{“(”}] * \mathfrak{e} * [\text{“)”}] \,|\, \mathfrak{e} \in \mathfrak{E}\} \\
\mathfrak{N} & := & N \cup N^* \\
N & := & \{\text{“0”}, \cdots, \text{“9”}\}
\end{array}
$$

What is the difference between pattern matching and tokenwize parsing?

2 `gen.jl` and `pattern.jl`

The *map form* of a pattern $\mathfrak{P}$ is a function $\Phi' : \mathfrak{P} \to L$, where $L \subseteq \mathcal{P}(\mathfrak{P})$ is a set of *labels*[1]. The *inverse map form* of a pattern $\mathfrak{P}$ is a function $\Phi : L \to \mathcal{P}(\mathfrak{P})$, where $\Phi(l) = \{p \in \mathfrak{P} \,|\, \Phi'(p) = l\}$ for $l \in L$.

Let $G := (\langle \mathscr{G} \rangle, *)$. Let $\triangle : [\Phi, L^2] \to \Phi$, where Φ is the inverse map form of a pattern $\mathfrak{P} \subseteq G^{<\omega}$ and $L = \mathrm{Dom}(\Phi) \subseteq G$, be defined by $\forall l \in \mathrm{Dom}(\Phi_1) \setminus \{l_1 * l_2\}$ $(\Phi_2(l) = \Phi_1(l))$ and $\Phi_2(l_1 * l_2) = \Phi_1(l_1 * l_2) \cup$

[1] Lables – or, *concepts* – are useful abstractions of patterns, which will be used later to optimize the complexity of solution-finding systems. The subsets $\mathfrak{T}, \mathfrak{F}, \mathfrak{N}$ and N in example 1.1.1 are examples of lables.

$\{[l_1, l_2]\}$ if $l_1 * l_2 \in \mathrm{Dom}(\Phi_1)$ and $\Phi_2(l_1 * l_2) = \{[l_1, l_2]\}$ if not, where $\Phi_2 = \triangle(\Phi_1, [l_1, l_2])$ and $l_1, l_2 \in \mathrm{Dom}(\Phi_1)$.